

Week 5

Understanding under Threat

Quantum Mechanics and Artificial Intelligence

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Today's plan

Part I: Quantum Mechanics

(lectures 1–2)

- The framework of QM: states, observables, dynamics
- The measurement problem and the Copenhagen myth
- Interpretations: GRW, Bohm, Everett, ...
- Complementarity: Bohr's response to the crisis
- Bell's theorem and the limits of causal understanding

Part II: Artificial Intelligence

(lecture 3)

- What genuine models explain: three examples from physics
- The double meaning of “model”
- Sullivan's argument: link uncertainty vs. implementation opacity
- A deeper challenge: is there an inside?

From last week, quantum mechanics is our recurring example of a theory where prediction and understanding seem to come apart.

1. Do you feel that you **understand** quantum mechanics? What would it even mean to understand it?
2. Is “shut up and calculate” a satisfying attitude toward a fundamental theory of nature?

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Two perspectives today

First-order Which interpretation of QM is correct? What is the theory *actually* saying about the world?

Second-order There has been no consensus on the foundations of QM for *100 years*. Should we be concerned? What does this tell us about science as a knowledge-producing enterprise?

Both questions bear directly on scientific understanding — which is why De Regt and Dieks chose quantum non-locality as their motivating example.

The foundational crisis: 1900–1927

- 1900: Planck introduces the quantum of action h
- 1905: Einstein explains the photoelectric effect using photons
- 1913: Bohr's model of the hydrogen atom
- 1925–26: Heisenberg's matrix mechanics; Schrödinger's wave mechanics
- 1927: Born's probability interpretation; Heisenberg's uncertainty principle; Bohr's complementarity

The early pioneers were acutely aware that they did *not* understand what was going on. Bohr thought our ability to understand the physical world was genuinely up for grabs.

Apparent resolution: the Copenhagen school

- By 1927 a working consensus formed around Bohr's school
- Bohr's key insight: the existence of the **quantum of action** calls for a *revision* of what it means to give an objective, unambiguous description of physical phenomena
 - An unambiguous description must specify the experimental arrangement; the conditions of observation become part of the description itself
 - Classical concepts remain indispensable for describing the apparatus, but cannot all be simultaneously applied to the quantum object
- This leads to **complementarity**: a structural feature of unambiguous description, not a limitation on knowledge
- This is a positive philosophical proposal, not a counsel of despair

Revival of the crisis

- 1952: Bohm publishes a working hidden-variable interpretation — shattering the apparent consensus
- 1957: Everett proposes the relative-state (many-worlds) interpretation
- 1964: Bell proves his non-locality theorem
- Today: active foundational debate; the cutting edge of physics proceeds without settling the foundations

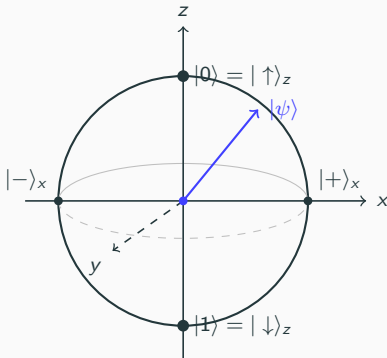
The second-order question: is 100 years without consensus acceptable? Does it reveal something deep about QM, or is it a sociological failure of physics?

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- The state of a quantum system is a unit vector $|\psi\rangle$ in a complex Hilbert space $\mathcal{H}$
- **Superposition principle:** if $|\psi\rangle$ and $|\phi\rangle$ are states, so is $\alpha|\psi\rangle + \beta|\phi\rangle$ (after normalization)
- For a **qubit** (two-level system): $\mathcal{H} = \mathbb{C}^2$
- Computational basis: $|0\rangle \equiv |\uparrow\rangle$, $|1\rangle \equiv |\downarrow\rangle$

The Bloch sphere



Every pure state of a qubit is a point on the sphere. Antipodal points are orthogonal.

Observables as operators

- Physical quantities: **Hermitian operators** $A = A^\dagger$
- Possible measurement outcomes: eigenvalues λ_i
- Eigenstates $|a_i\rangle$: states with definite value λ_i

Spin operators (Pauli matrices $\times \hbar/2$):

$$S_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad S_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad S_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

Superpositions and the puzzle

If the system is in $|\psi\rangle = \alpha|a_1\rangle + \beta|a_2\rangle$, observable A has *no definite value* — yet measurement always yields a definite outcome λ_1 or λ_2 .

The puzzle: what is the system doing between preparation and measurement? Does the question even make sense?

This tension is the engine of every interpretation of QM.

Uncertainty relations

Non-commuting observables cannot both have definite values:

$$[S_y, S_z] = i\hbar S_x \implies \Delta S_y \cdot \Delta S_z \geq \frac{\hbar}{2} |\langle S_x \rangle|$$

Spin-z eigenstate decomposed in spin-y basis:

$$|\uparrow\rangle_z = \frac{1}{\sqrt{2}}|\uparrow\rangle_y + \frac{1}{\sqrt{2}}|\downarrow\rangle_y$$

so $\Delta S_y = \hbar/2$ is maximal when S_z is definite.

Conceptually: **Fourier analysis** — narrow in one basis, spread in the conjugate basis.

Composite systems and entanglement

Composite systems: $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$, with
 $x \otimes (y + z) = (x \otimes y) + (x \otimes z)$.

Not all states are product states. The **singlet state**

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

cannot be written $|\psi\rangle_A \otimes |\phi\rangle_B$: it is **entangled**.

Discussion (2 min): what features does a system have if it is entangled with another? What does it mean for its “state”?

Linear dynamics and the Born rule

Time evolution: $i\hbar \frac{d}{dt}|\psi\rangle = H|\psi\rangle$, solution $|\psi(t)\rangle = U(t)|\psi(0)\rangle$ (**unitary**).

Crucially, U is **linear**: $U(\alpha|\psi\rangle + \beta|\phi\rangle) = \alpha U|\psi\rangle + \beta U|\phi\rangle$. This is exactly what generates the measurement problem.

Born rule:

$$\Pr_{\psi}(A = \lambda) = \langle \psi, E^A(\lambda) \psi \rangle$$

For $|\psi\rangle = \sum_i c_i |a_i\rangle$: $\Pr_{\psi}(A = \lambda_i) = |c_i|^2$

Discussion: do we *understand* QM if it is just a recipe for computing these probabilities?

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Definition and derivation

Ideal measurement of A : $|a_i\rangle \otimes |r\rangle \longrightarrow |a_i\rangle \otimes |y_i\rangle$ (each eigenstate paired with a distinct pointer state — unitary).

Now suppose $|\psi\rangle = \sum_i c_i |a_i\rangle$. By **linearity**:

$$\left(\sum_i c_i |a_i\rangle \right) \otimes |r\rangle \longrightarrow \sum_i c_i (|a_i\rangle \otimes |y_i\rangle)$$

The meter ends in an **entangled superposition** — no definite reading. But we *always* observe a definite outcome.

Contradiction.

The collapse postulate

Standard “solution”: $\sum_i c_i |a_i\rangle \longrightarrow |a_k\rangle$ with probability $|c_k|^2$.

Problems:

- Violates unitarity and linearity — the two most fundamental features of the theory
- Creates a two-dynamics theory: Schrödinger evolution *and* collapse
- When does collapse occur? Where is the “Heisenberg cut”?
- The theory gives no definition of “measurement”

The Copenhagen myth

- Von Neumann (1932) first axiomatized collapse in *Grundlagen der Quantenmechanik*; by the 1950s it was standard textbook orthodoxy
- The label **Copenhagen interpretation** was applied retroactively — Bohr never used it
- In reality there was no single Copenhagen view: Bohr, Heisenberg, Born, Pauli all held different positions
- **HH:** Bohr never claimed the measurement problem is *solved* by collapse — his actual response is philosophically much deeper, and has been systematically misread

Decoherence: not a solution

Decoherence: environmental interaction entangles the system rapidly; the *reduced* state approaches a mixture.

Why this does not solve the measurement problem:

1. The reduced state is not *uniquely* interpretable as a mixture over definite pointer readings
2. The global state (system + environment) is still a superposition — decoherence is a local approximation, not a collapse

Decoherence explains why interference is hard to observe macroscopically. It does not explain why we observe *one* outcome.

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Four strategies

1. **Reject unitary dynamics:** add stochastic collapse (*GRW*)
2. **Add hidden variables:** supplement QM with additional ontology (*Bohm*)
3. **Reject uniqueness of outcomes:** all branches are real (*Everett*)
4. **Revise the ontological picture:** resist the demand for a classical reality behind QM (*relational, QBism, pragmatic*)

GRW (Ghirardi, Rimini, Weber 1986): add random spontaneous localization events. Free parameters tuned so macroscopic objects localize (no pointer superpositions) while microscopic systems remain unitary. Open: ontology? Relativistic extension?

Bohm (1952): both wavefunction and particle positions are real; the wave guides particles. Particles always have definite positions.

- Guidance equation requires a preferred frame — inconsistent with special relativity
- Violates the spirit of momentum conservation in relativistic interactions

Everett and less ontological interpretations

Everett (1957): unitary dynamics is complete; all branches of the wavefunction are real.

- *Probability*: deterministic theory — where do the Born-rule probabilities come from?
- *Preferred basis*: what selects the “world” branches?
- No-hidden-variables theorems imply the branches cannot themselves be classical

Less ontological (Rovelli, Fuchs–Schack, Healey): resist asking “what is the wavefunction in itself?”

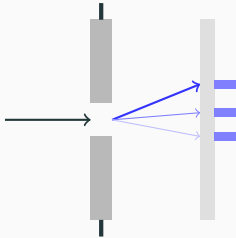
HH: these approaches are philosophically sophisticated but may relocate rather than dissolve the problem.

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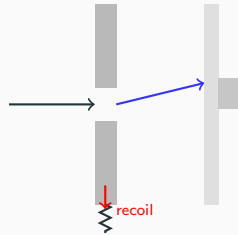
Bohr's single-slit thought experiment

A: bolted screen



position known $\Rightarrow$ wave picture $\Rightarrow$
interference

B: free screen



momentum known $\Rightarrow$ particle
picture $\Rightarrow$ **no interference**

Complementarity and visualizability

Two arrangements are **complementary** when they are mutually exclusive and each exhaustive within its domain.

Spacetime description position, trajectory, “which path” —
visualizable (anskuelig) picture

Causal description momentum, energy, conservation laws —
dynamical picture

“The viewpoint of complementarity presents itself as a rational generalization of the very ideal of causality.” (Bohr 1948)

This connects directly to De Regt & Dieks: Bohr is *not* saying we cannot understand quantum phenomena. He is proposing a new kind of understanding that does not require a single classical picture.

Bohr's deeper point

- The quantum of action requires a revision of what counts as an **objective description**:
 - Objectivity does not disappear — but it now requires specifying the experimental conditions as part of any unambiguous statement about a quantum phenomenon
 - We cannot separate “how Nature is” from “the experimental arrangement in which it is observed”
- This is not about abandoning objectivity but about *reconceiving* it for a domain where the interaction between object and instrument cannot be neglected or controlled
- The misreading of Bohr as “shut up and calculate” has distorted the foundations debate for 70 years
- **HH:** Bohr is working in a tradition running from Kant through Høffding: the conditions of unambiguous description are themselves a philosophical achievement

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EPR, von Neumann, and Kochen–Specker

EPR (1935): QM is incomplete — quantum correlations must be explained by predetermined hidden variables λ .

Von Neumann (1932): no hidden-variable theory can reproduce QM given additivity $\mathbb{E}_\lambda(A + B) = \mathbb{E}_\lambda(A) + \mathbb{E}_\lambda(B)$ for non-commuting A, B . Bell (1966): this assumption is physically unmotivated.

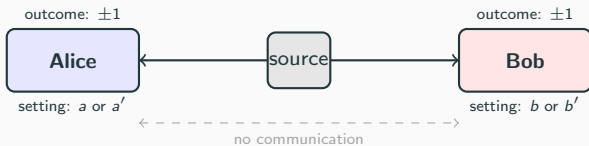
Kochen–Specker (1967): no assignment $v(A) \in \text{spec}(A)$ to all observables can satisfy functional consistency $v(f(A)) = f(v(A))$ and the sum rule for commuting pairs. **Value definiteness** is impossible.

Kochen–Specker is more compelling than von Neumann: the assumptions are operationally motivated.

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The CHSH scenario



The entangled pair is measured far apart. Each party independently chooses a setting and records a ± 1 outcome.

Local hidden variables and the CHSH inequality

Suppose outcomes are determined by $\lambda \sim \rho(\lambda)$, with **reality** (A, B definite) and **locality** (Alice independent of Bob's setting):

$$E(a, b) = \int A(a, \lambda) B(b, \lambda) \rho(\lambda) d\lambda$$

Under local hidden variables:

$$|E(a, b) - E(a, b') + E(a', b) + E(a', b')| \leq 2$$

Proof sketch: for each λ , the combination $A(a)[B(b) - B(b')] + A(a')[B(b) + B(b')]$ takes values in $\{-2, 0, +2\}$.

Quantum violation

For $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$: $E(a, b) = -\cos(\theta_a - \theta_b)$

Choose $a = 0^\circ$, $a' = 90^\circ$, $b = 45^\circ$, $b' = 135^\circ$:

$$E(a, b) = -\cos(-45^\circ) = -\frac{1}{\sqrt{2}}$$

$$E(a, b') = -\cos(-135^\circ) = +\frac{1}{\sqrt{2}}$$

$$E(a', b) = -\cos(45^\circ) = -\frac{1}{\sqrt{2}}$$

$$E(a', b') = -\cos(-45^\circ) = -\frac{1}{\sqrt{2}}$$

$$S = -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = -\frac{4}{\sqrt{2}} = -2\sqrt{2} \quad \Rightarrow$$

$$|S| = 2\sqrt{2} \approx 2.83$$

The Jarrett analysis and what Bell means

Jarrett (1984): local hidden variables = two conditions:

PI (parameter independence) Alice's outcome does not depend on Bob's *setting*

OI (outcome independence) Alice's outcome does not depend on Bob's *outcome*

Bell violation $\Leftrightarrow$ at least one of PI or OI fails. Bohm/Everett: OI fails. Retrocausal models: PI fails.

HH's focus: Bell's theorem shows a fundamental limitation on **local causal explanation** of quantum correlations. Does this mean we *cannot understand* them?

De Regt & Dieks: understanding does not require causal explanation.
Bohr: we need a new conceptual framework, not a new causal story.

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- QM works to extraordinary precision (electron g -factor: 12 significant figures)
- Yet the theory is arguably *inconsistent as stated* (the measurement problem), and there is no consensus on how to resolve it
- Each interpretation pays a significant price: non-locality, many worlds, anti-realism, new fundamental dynamics
- **HH:** this is a genuine crisis — not merely a philosophical puzzle we can comfortably set aside

- Some physicists say: “QM works, so who cares if we don’t understand it?”
- If we accept this for QM, we have weakened our ground against the parallel argument:
“AlphaFold predicts protein structures better than any human theory — so who cares if we don’t understand it?”
- **HH:** I do not think we should be content with this. The goal of science is not mere prediction but understanding. QM is one thing; the threat from AI is another.

Part II

Artificial Intelligence and Scientific Understanding

AlphaFold (2022) predicted the 3D structure of virtually every known protein — a problem biologists had worked on for 50 years. AlphaFold is a deep neural network trained on known protein structures.

1. Did AlphaFold help us **understand** protein folding? Or did it only help us **predict** protein structures?
2. Is there a meaningful difference between a neural network that predicts and a model that explains?

9. AI in Science

10. What Models Explain

11. Sullivan's Argument

12. A Deeper Challenge

Two perspectives on AI in science

Theoretical Videnskabsteori has traditionally been focused on *scientists* as cognitive agents. These agents have always used tools to increase understanding. But now we have tools that seem borderline intelligent themselves. Does that change our understanding of the task of science?

Practical Science produces (at least) two kinds of goods:

- **Intrinsic:** The goods it gives back to scientists and their lives — helping them fulfill their human potential
- **Extrinsic:** The goods it produces for our material lives and the economy

Use of AI threatens to make the extrinsic good completely *enådig* (one-sided). What do we want science to be? It is partly up to us.

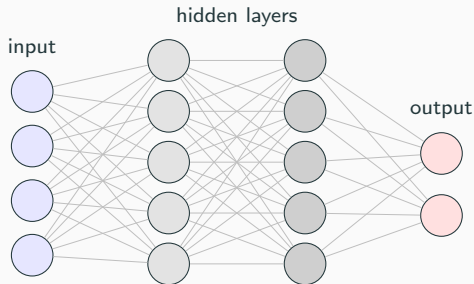
A fact about AI: opacity

Deep Neural Networks (DNNs) — including the transformer models behind AlphaFold and large language models — share one crucial feature:

Even the experts who build them do not know how they work

- There is a fundamental **mismatch** between human reasoning and the path a DNN takes from inputs to outputs
- A DNN learns by adjusting billions of numerical weights through an optimization process; the resulting weights are not human-readable
- Joel Dudley (co-creator of the Deep Patient model): “We can build these models, but we don’t know how they work”

What a DNN looks like inside



each edge carries a weight $w_{ij} \in \mathbb{R}$, learned by gradient descent — not designed to represent any physical quantity

The “knowledge” lives in the weights. These are the output of gradient descent over training data — not designed to represent any physical mechanism or causal structure.

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What makes a model genuinely explanatory?

Before turning to Sullivan, let us recall what it looks like when physics provides genuine understanding.

A **model** in the philosophy-of-science sense:

- Has *interpretable content* — it says something about what is going on in the world
- States a mechanism, a structure, a *reason why*
- Scientists can read off qualitative consequences without full calculation (De Regt & Dieks: CIT)
- Transfers understanding across domains

Three examples.

Example 1: Kinetic theory of gases

Model: gas = N molecules of mass m moving randomly; pressure arises from momentum transfer to container walls; temperature $\propto$ mean kinetic energy $\langle \frac{1}{2}mv^2 \rangle$

Explains: why hydrogen effuses $4\times$ faster than oxygen at the same temperature:

$$\langle \frac{1}{2}mv^2 \rangle \text{ equal for both} \Rightarrow v \propto m^{-1/2} \Rightarrow \frac{v_{\text{H}_2}}{v_{\text{O}_2}} = \sqrt{\frac{m_{\text{O}_2}}{m_{\text{H}_2}}} = \sqrt{\frac{32}{2}} = 4$$

You can see why the answer is 4 without computing anything. The billiard-ball picture carries the understanding.

Example 2: Schwarzschild metric and light deflection

Model: the unique spherically-symmetric vacuum solution (Birkhoff) to Einstein's equations:

$$ds^2 = -\left(1 - \frac{2GM}{c^2 r}\right) c^2 dt^2 + \left(1 - \frac{2GM}{c^2 r}\right)^{-1} dr^2 + r^2 d\Omega^2$$

Explains: deflection of light grazing the Sun,
 $\delta = 4GM/c^2 b \approx 1.75''$ — *exactly twice* the Newtonian prediction.

Once you have internalized the geometry, you can see why the factor of 2 arises: spatial curvature (absent in Newton) contributes an equal amount to the deflection as the time-component does. You see it; you do not just calculate it.

Example 3: BCS theory of superconductivity

Model: phonon-mediated attraction pairs electrons of opposite momenta/spins into **Cooper pairs**; these condense into a coherent macroscopic state with an energy gap Δ .

Explains — qualitatively, without calculation:

- **Zero resistance:** the minimum excitation costs 2Δ ; for small voltages there are no available scattering states — current flows without dissipation
- **Meissner effect:** a penetrating magnetic field would disrupt the condensate; the material expels it
- **Breakdown at T_c :** thermal fluctuations supply $k_B T \gtrsim \Delta$, breaking the pairs

A physicist who has internalized BCS can answer *why* mercury loses resistance at 4.2 K — not merely predict that it does.

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“Are scientists trading understanding for some other epistemic or pragmatic good when they choose an opaque and complex machine learning model?”

(Sullivan 2022, p. 110)

Discuss: do you think scientists are making such a trade? Is that acceptable?

The double meaning of “model”

One terminological trap in Sullivan’s article: the word **model** does double duty.

Sense 1 (scientific model) A representation with *interpretable content*: states a mechanism; has an inside that scientists can inhabit; satisfies CIT. A Sense 1 model *says something about what is going on in the world*.
Examples: kinetic theory, Schwarzschild metric, BCS

Sense 2 (ML model) A parameterized function *fitted to data by optimization*: weights are the output of gradient descent, not designed to represent anything. A Sense 2 model is *the result of a fitting procedure, not of theorizing*.
Examples: AlphaFold, GPT, Deep Patient

Sullivan's main claim

Sullivan's answer to her own question: it is *not* the complexity or opacity of a DNN that limits the understanding it can provide.

The real culprit is **link uncertainty**:

"It is not implementation black-boxing that gets in the way of understanding, it is link uncertainty."

(Sullivan 2022, p. 116)

- Black-boxing implementation details does *not* undermine understanding in general — this is commonplace in science
- What blocks understanding is the absence of scientific evidence linking the model's outputs to real causal structure in the world

Discuss: do you think Sullivan is right about this? Can you give examples of acceptable and unacceptable black-boxing from physics?

The Schelling analogy

Sullivan's central example: Schelling's **checkerboard model** of residential segregation.

- Simple simulation: two types of agent, each moves if $> 70\%$ of neighbours differ; result: a segregated board
- Provides *how-possibly* understanding when the semantic mapping (coin = agent, move = choice) is read off
- Becomes *how-actually* understanding only when empirical evidence connects the model to real populations

Sullivan's claim: the same structure applies to DNNs.

Implementation opacity is not the obstacle — link uncertainty is.

HH: But notice the key difference. With Schelling, the semantic mapping was *built in by the modeller from the start*. With Deep Patient, there is no such mapping. The very question of what the nodes and weights *refer to* is unanswered — and it is not obvious it is even askable.⁵⁵ / 61

Sullivan's optimism and a challenge

"Once the link uncertainty is resolved, the deep patient model is able to explain and enable understanding of disease development."

(Sullivan 2022, p. 125)

HH: How would the link uncertainty ever be resolved in this case?

- With Schelling: we look for evidence that individual preferences actually drive segregation — this is a clearly statable empirical question
- With Deep Patient: what physical objects or processes would the learned weights be *linked to*?
- The weights emerged from optimization over medical records. There is no design that maps them onto causal factors in disease. The link question cannot be stated in the same form.

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The internal/external distinction (Sullivan's blind spot?)

When a physicist derives a result, her brain is — at the physical level — just as opaque as a DNN.

If you traced the neural mechanisms, you would not see the reasoning. To understand why she gets the right answers, you have to get inside the **structure of the reasoning itself**.

Leibniz's mill (*Monadologie*, §17): even if the brain were enlarged to the size of a mill, and you could walk around inspecting all its machinery, you would never encounter *thought* anywhere in the mechanism.

Sullivan asks: how opaque is the implementation?

But there is a prior question: **is there an internal perspective to be had at all?**

- With the physicist: yes — her reasoning has a logical

Van Fraassen: a model must be a “way the world could be”

Van Fraassen is the arch empiricist: he does not ask that our theories be *true*, only *empirically adequate*.

Yet even van Fraassen would not be satisfied with Deep Patient as a final scientific result.

His **constructive empiricism** is a *semantic* empiricism, not an eliminativist one. A scientific theory presents models — structures interpretable as *ways the world might be*. The scientist may be agnostic about truth, but she must be able to say: this is a *possible description of reality*.

A DNN does not present a way the world could be. It presents a function from inputs to outputs. The question of truth or falsity about disease mechanisms does not even arise.

For reflection

1. Should scientists rest content with having something as predictive as Deep Patient? Or does science have a further goal that remains unfulfilled?
2. Does it matter whether we are talking about *applied* science (predicting disease, protein structure) vs. *fundamental* science (understanding the laws of nature)?
3. We saw in Part I that QM proceeds without consensus on what it means. Does this weaken or strengthen the case for demanding understanding from AI?
4. Intimation: the problem of understanding another human being — and why that seems different from understanding a DNN — points toward questions that philosophy has been wrestling with for centuries.

